

Answer Key: Political Polarization and Right-wing Populism

Exercise

Part 1: Hypotheses

Null Hypothesis (H_0): Exposure to polarizing political content on social media has no effect on support for right-wing populist candidates. (The difference in support rates between the groups is zero.)

Alternative Hypothesis (H_a): Exposure to polarizing political content on social media increases support for right-wing populist candidates. (The support rate is higher in the polarizing content group than in the control group.)

Part 2: Data Analysis

Group	Support Candidate	Don't Support	Total	Support Rate
Polarizing Content Group	42	58	100	42%
Control Group	28	72	100	28%

Calculations:

1. Support rate for Polarizing Content Group: $\frac{42}{100} = 42\%$
2. Support rate for Control Group: $\frac{28}{100} = 28\%$
3. Difference in support rates: $42\% - 28\% = 14\%$

Decision: Since the difference (14%) is greater than 10%, we reject the null hypothesis.

Conclusion: The data suggests that exposure to polarizing political content on social media significantly increases support for right-wing populist candidates.

Part 3: Error Types & Sampling Error

1. Type I Error in this context:

A Type I error would occur if we conclude that exposure to polarizing content increases support for right-wing populist candidates when, in reality, it has no effect. This means rejecting the null hypothesis when it is actually true.

2. Type II Error in this context:

A Type II error would occur if we conclude that exposure to polarizing content has no effect on support for right-wing populist candidates when, in reality, it does increase support. This means failing to reject the null hypothesis when it is actually false.

3. Effects of sampling error:

Sampling error could lead to observed differences between the groups that don't accurately reflect the true relationship in the population. With a relatively small sample size ($n=100$ per group), random variation might exaggerate or minimize the true effect of polarizing content. A larger sample size would reduce sampling error and provide more reliable results.

4. Problems with limited sample demographics:

If the research only surveyed voters from a particular age group or geographic region, the results may not generalize to the broader population. For example:

- If only surveying young voters, the study might overestimate the effect since younger people might be more influenced by social media content.
- If only surveying voters from urban areas, we might miss different patterns of response in rural regions.
- Surveying only highly educated voters might underestimate effects, as education level could moderate susceptibility to polarizing content.